

Bonus Task for 5 Points
Deadline: December 13, 2025

You can submit your bonus task to the assistants according to the following schedule:

- **Nurbek Mauletkhan:** 13.12, 14:00-16:00 - **344**
- **Nurasyl Sakaev:** 13.12, 09:00-11:00 - **438**
- **Anuar Ospanov:** 13.12, 12:00-13:00 - **451** and 14:00–15:00 - **365**

1. Choose a Topic

Select one of the following topics:

- Library Management System
- Car Rental Service
- Flight Booking System
- University Course Registration System
- Restaurant Reservation & Ordering System.

2. ER Diagram

Design an ER diagram with at least six entities. Include attributes with appropriate data types, relationships (one-to-one, one-to-many, many-to-many), primary keys, and foreign keys. Export the ER diagram as a PDF.

3. Database and Tables

Create a new database and all tables from your ER diagram using PostgreSQL in DataGrip. Include the following constraints: primary key, foreign key, unique, not null, and at least three meaningful check constraints (for example, CHECK (price > 0), CHECK (age >= 18), or CHECK (status IN ('pending','confirmed','cancelled'))).

4. Insert Sample Data

Insert your own realistic data into the tables. Include at least 20 rows per table. Ensure that foreign key relationships are correct and that dates, prices, emails, and IDs are valid.

5. Create Indexes

Create at least five indexes: a simple B-tree index, a composite (multi-column) index, a unique index, a functional index (e.g., using LOWER() or DATE()), and a partial index.

6. SQL Queries

Write at least 12 meaningful SELECT queries, including: INNER JOIN and LEFT JOIN, GROUP BY with aggregates, subqueries in FROM or WHERE, ORDER BY and LIMIT, queries that use indexes, and queries with filters and conditions.

Example tasks: find the most rented car, top 5 books with the highest reservations, flights with the most passengers, courses with no available seats, or the most popular dish.

7. Roles and Users

Create three roles: admin, manager, and viewer. Assign specific permissions to each role and create at least two users with assigned roles.

8. Additional Component

Choose one additional component: a transaction, a function, or a view.

- **Transaction:** example - booking a flight, renting a car, borrowing a book, or placing a restaurant order. It should include BEGIN, multiple operations, and COMMIT or ROLLBACK.
- **Function:** example - calculate order total, available seats, overdue fees, or rental cost.

- **View:** example - available cars, active reservations, top booked flights, or students with overloaded schedules.

9. Data Analysis

After inserting all data, perform analysis based on your dataset. Include: the top 3 most popular items, the least used items, total revenue, users/customers with the highest activity, at least 3 data anomalies with explanations of how normalization solved them, and 2 business insights based on your data (e.g., evening flights are most popular or business customers rent cars more).

10. Project Defense

During the defense, show the ER diagram, table structure, execute all SQL queries, explain constraints, indexes, roles, and the additional component (transaction/function/view), and present the results of the data analysis.

During the defense, the instructor may ask the student to perform quick SQL tasks directly in the console.

! Submission Instructions

The entire project, including all diagrams, queries, explanations, and analysis, must be compiled into **a single PDF file** and submitted via **MS Teams before the defense**.